GEORGIA INSTITUTE OF TECHNOLOGY
SCHOOLS OF CS AND ECE
ECE 4100/6100, CS 4290/6290 (CONTE)
REVIEW FOR FINAL EXAM
FINAL: THURSDAY, MAY 6TH, 8:00 - 10:50 AM EDT (UTC-4), ON LINE

5. **ALL TOPICS FROM EXAMS 2 AND 2**
6. **Instruction-Level Parallelism (N5 part II)**
   a) VLIW vs. Superscalar: how they differ
   b) Scheduling code for a VLIW
   c) Instruction encoding tradeoffs: tail bit, pause field
      i) Global acyclic scheduling:
         a) Trace scheduling
         b) Superblocks
         c) Hyperblocks
         d) Treegions
         e) Advantages/disadv. of each scheme
   d) Loop unrolling
   e) Software pipelining:
      i) What are the prologue, kernel and epilogue
      ii) Rotating registers
      iii) Iterative modulo scheduling: Finding MII, ResMII and RecMII, kernel, prologue and epilogue
7. **Multiprocessors (N6)**
      i) Classes of multiprocessors:(1) Centralized shared memory (UMA) (2) Distributed shared memory (NUMA) (3) Distributed memory with message passing (no shared memory)
   b) Cache coherence
      i) Definition and 3 properties
      ii)Shared caches: advantages and disadvantages
      iii) Snooping
         a) Processor requests: GetM, GetS
         b) Actions on the bus: GetM, GetS, Abort or Intervene
         c) MSI protocol
      iv) Meaning of M, O, E, S, I and F states
      v) Cache to cache transfers
      vi) Write-update vs. Write-invalidate protocols
      vii) False sharing
      viii) Directory-based coherence
         a) Presence bits (Censier and Featrier), MSI version (in notes)
         b) Limited-pointer directory schemes (e.g., Dir-i-B)
         c) Multicore Lowest-Level Private vs. Lowest-Level Shared caches
7. **Synchronization and Consistency (N7)**
   a) Synchronization
      i) Mutex synchronization
         a) Implementing Getlock()
         b) Implementing test and set (TS) – coherence protocol problems with TS
         c) Test and Test and Set (TTS) – and why it helps coherence traffic
         d) Fetch and Add
         e) Load-Linked / Store Conditional
      ii) Barrier synchronization
   b) Transactional memory
   c) Memory consistency
      i) The problem
      ii) Strict consistency
      iii) Sequential consistency
      iv) Relaxed consistency example: Weak consistency

Plus: All material on Review Sheets for Exams 1 & 2, all Homework (including practice HW), all Projects